

The Battery Industry — Global, China, and the India Cell Bet

A framework for Exide (EXID IN) and Amara Raja (ARENMOB IN)

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TL;DR

- **The global battery market is structurally growing but cyclically oversupplied, and China owns it.** Demand crossed c.1 TWh in 2024 and heads toward c.3.9–4.7 TWh by 2030, yet nameplate capacity already exceeds 4 TWh (>80% Chinese), BloombergNEF's 2025 survey put pack prices at a record-low **\$108/kWh** (–8% YoY), and CATL+BYD alone installed 659.5 GWh — 55.6% — of the 1,187 GWh global EV battery market in 2025. Building cells outside China against this backdrop is a margin trap unless protected by policy.
- **India is a real demand story but an unproven supply story.** The ACC PLI scheme has delivered just **1.4 GWh of a 50 GWh target (2.8%)** four years in — entirely by Ola Electric, with zero incentives disbursed — and India remains c.100% import-dependent on cells. BESS is the fastest, most credible pull (CEA needs 236 GWh of BESS by FY32 vs a c.68 GWh pipeline today), but tariffs have collapsed to **Rs1.48 lakh/MW/month**, and c.290 GWh of announced cell capacity vs credible demand points to eventual overcapacity, not comfortable profitability.
- **On the two stocks, we favour lead-acid cash generation over lithium optionality — and so does the street.** Both Exide (12 GWh, ~Rs6,000 cr) and Amara Raja (16 GWh, Rs9,500 cr) [spglobal](#) are funding lithium out of lead-acid cash flows, both are burning money pre-revenue (EESL lost Rs248 cr in FY26), [Whalesbook](#) and the street ascribes near-zero premium above invested capital to the cell business (Exide lithium at Rs59–60/share; Amara Raja at Rs169/share, 1x book). The cell bet creates value only if import duties arrive, offtake is locked, and utilisation clears breakeven — none of which is yet true.

Key Findings

1. **Structural vs cyclical.** The *structural* trend — electrification of mobility and grid storage driving multi-TWh battery demand — is intact and accelerating in stationary storage. The *cyclical* reality of 2024–26 is brutal overcapacity, collapsing cell/pack prices, and metal-price deflation. Investors are conflating the two: a great demand chart does not make a great cell-manufacturing investment.
 2. **China's dominance is now a chokepoint, not just a cost edge.** China holds >80% of cell capacity, c.85–90%+ of cathode/anode/electrolyte/separator processing, >90% of graphite processing, and in October 2025 extended export controls to LFP cathode tech, lithium-processing equipment, and battery materials. Anyone building non-Chinese cells depends on Chinese equipment, materials, and licensed technology — exactly the dependency both Indian players have signed up for (Exide–SVOLT, Amara Raja–Gotion).
 3. **LFP has won the volume war; sodium-ion is the next deflationary shock; solid-state is a premium 2027–2030+ story.** LFP is now c.81% of Chinese installations. CATL's Naxtra sodium-ion (up to 175 Wh/kg) enters mass production in 2026 with a 60 GWh storage order — a genuine threat to entry-level and stationary LFP. Solid-state (Toyota 2027–28) is real but small and premium for a decade.
 4. **India demand is led by 2W/3W and BESS, not cars.** EV penetration was c.7.3% of vehicles in FY26 (2W/3W ~87% of EV volumes); electric cars remain low single digits. The credible, contracted pull is grid BESS: 130+ GWh tendered in 2025; CEA targets 236 GWh BESS by FY32.
 5. **The India capex cycle is heading toward oversupply — after a period of import dependence.** c.290 GWh of gigafactory intentions by 2030 against realistic domestic cell demand of perhaps 100–150 GWh implies China's/Europe's fate if all builds. Near term the risk is the opposite — undersupply and c.100% import reliance, with Nuvama flagging a c.210 GWh shortfall vs India's 2032 BESS requirement.
 6. **For Exide and Amara Raja, lead-acid is the cash cow and the reason to own the stocks; lithium is a call option the market prices at roughly cost.** Exide core lead-acid EBITDA margin is c.11–12%, zero debt; Amara Raja LAB EBIT margin recovered to 13.2% in Q4FY26. CompoundingAI The lithium subsidiaries are loss-making and capital-hungry.
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Details

1. Global battery demand — structurally up, forecasts diverge on magnitude

Global lithium-ion demand crossed c.1 TWh in 2024 and continues to compound, driven

overwhelmingly by mobility with stationary storage the fastest-growing add. Forecasts diverge widely on 2030 magnitude — a spread investors must respect.

Table 1 — Global Li-ion battery demand forecasts to 2030 (GWh; actuals vs estimates)

House	2030 total demand	Notes	A/E
McKinsey (2023)	c.4,700 GWh	Mobility c.4,300 GWh of total	E
Bain (2024)	c.4,100 GWh	"quadruple 2023–2030"	E
McKinsey (2025, base)	c.3,910 GWh (from c.1,970 GWh in 2025)	More conservative revision	E
IEA GEVO STEPS	EV demand ×4.5 vs 2023 by 2030	+ c.400 GWh stationary (STEPS)	E
RMI	5,500–8,000 GWh ("fast or faster")	Bull-case outlier	E
Global EV installs 2024	901.4 GWh (SNE Research)	Crossed 1 TWh incl. non-EV	A
Global EV installs 2025	1,187 GWh (SNE Research)	+31.7% YoY	A

Bottom line: demand roughly quadruples this decade in every serious scenario; the debate is c.3.9 vs c.4.7 TWh. Stationary storage is the swing factor — McKinsey sees total installed BESS reaching 500–700 GWh by 2030 from c.200 GWh end-2025.

2. Overcapacity and the price collapse — the cyclical reality

Nameplate capacity already exceeds demand. Global nameplate reached >4 TWh end-2025 (IEA), up c.30% YoY, with China >80% of the total and the EU/US 6–7% each. McKinsey's base case shows Chinese oversupply persisting through 2030 even if all announced capacity materialises; Chinese utilisation runs c.50–55% against c.4,800 GWh planned vs c.1,000 GWh domestic demand.

Prices hit record lows. BNEF's 2025 Lithium-Ion Battery Price Survey (released 9 Dec 2025) put volume-weighted **pack prices at \$108/kWh** (-8% YoY, -93% vs 2010). LFP packs averaged \$81/kWh vs \$128/kWh for NMC; **stationary-storage packs fell 45% to \$70/kWh**, the cheapest segment for the first time. The lowest observed LFP cell/pack prices were \$36/\$50/kWh in China. [BloombergNEF](#) China's average pack was \$84/kWh vs North America +44% and Europe +56% [ESS News](#) — the geographic premium that defines

the “build outside China” problem. Critically, BNEF noted metal prices *rose* in 2025 yet packs still fell: the price driver is now overcapacity and LFP mix, not commodities.

Table 2 — BNEF 2025 battery prices (\$/kWh, volume-weighted)

Metric	2024	2025	A/E
Pack (all)	~\$115	\$108	A
Cell (all)	~\$78	\$74	A
BEV pack	\$97	\$99	A
Stationary pack	~\$127	\$70	A
LFP pack	—	\$81	A
NMC pack	—	\$128	A
China avg pack	—	\$84	A

3. Competitive landscape — a Chinese duopoly with a Korean/Japanese tail

Table 3 — Global EV battery installations, 2025 (SNE Research)

Rank	Maker	Country	2025 share	A/E
1	CATL	China	39.2% (464.7 GWh)	A
2	BYD	China	16.4% (194.8 GWh)	A
3	LG Energy Solution	Korea	c.9.3%	A
4	CALB	China	c.4.9%	A
5	Gotion High-tech	China	c.4.3%	A
6	SK On	Korea	c.3.9%	A
7	Panasonic	Japan	c.3.7%	A
8–10	EVE / SVOLT / Samsung SDI	China/Korea	c.2.6% each	A

CATL + BYD installed 659.5 GWh — 55.6% — of the 1,187 GWh installed in 2025. Six of the top ten are Chinese (c.69% combined). Note the two Indian technology partners sit here:

Gotion (5th, Amara Raja’s LFP partner) and **SVOLT (10th, Exide’s partner)** — the Indians

are licensing from mid-tier Chinese players, not the leaders.

Northvolt is the cautionary tale. Europe's flagship raised >\$15 bn, secured c.\$50 bn of orders, and reached only c.1 GWh vs a 16 GWh target before filing for bankruptcy in March 2025. The lesson: a Chinese gigafactory costs c.€1.5 bn and would cost c.€3 bn to replicate outside China; capital, yield curves, and ramp risk destroy Western greenfields absent deep protection. This is the single most relevant global precedent for the India cell bet.

4. Chemistry evolution — LFP now, sodium next, solid-state later

Table 4 — Chemistry scorecard and adoption outlook

Chemistry	Energy density	Cost	Cycle life	Where it wins	Timeline
LFP	150–180 Wh/kg	Lowest lithium (\$81/kWh pack)	High	Mass EVs, 2W/3W, BESS	Now — dominant (c.81% China)
NMC (high-Ni)	250–300 Wh/kg	High (\$128/kWh pack)	Moderate	Premium/long-range EVs	Now, but share ceding to LFP
Sodium-ion	up to 175 Wh/kg (CATL Naxtra)	Below LFP at scale (~\$40/kWh cell target)	>10,000 cycles	Entry EVs, 2W/3W, cold-climate, BESS	Mass production Q2 2026
LMFP/LFMP	+15–20% vs LFP	Near-LFP	High	Mid-range EVs	Ramping
Solid-state (sulfide)	400–500 Wh/kg	Very high initially	TBD	Premium/flagship EVs	2027–28 pilot; mass 2030–33
Lead-acid / lead-carbon	30–50 Wh/kg	Very low	Low–moderate	SLI, UPS/inverter, telecom, standby	Persistent replacement franchise

Sodium-ion is the wildcard for India. CATL unveiled Naxtra (up to 175 Wh/kg) at its Super Tech Day on 21 April 2025, targets mass production in Q2 2026 (first passenger vehicle, the

Changan Nevo A06), and signed a three-year agreement to ship **60 GWh of sodium-ion batteries to Beijing HyperStrong** for energy storage. For Indian entry-level 2W/3W and stationary storage — precisely the segments Exide and Amara Raja target first — sodium could undercut the LFP economics both are building toward. Caveat: today's low lithium prices (\$10,163–10,233/t battery-grade carbonate in China, Oct 2025, -88% from the Nov-2022 peak) have blunted sodium's near-term cost edge.

Solid-state: Toyota's official roadmap is 2027–28 for first commercial (Lexus/premium), with only "a few hundred tons" — tens of thousands of vehicles — in early years; genuine mass-market is 2030–33. Treat announced dates as aspirational, not banked.

5. Raw materials and the China chokepoint

Lithium carbonate has round-tripped: from CNY 590–605k/t (Nov 2022 peak) down to c.\$10,200/t (Oct 2025), with a Q4-2025/early-2026 bounce toward CNY 155,000/t. **China's control is the strategic risk.** China processes >90% of graphite, c.85–90%+ of cathode/anode/electrolyte/separator, and two-thirds of lithium and cobalt refining. On **9 October 2025** China extended export controls to lithium-ion batteries, LFP CAM, lithium-processing technology and equipment, and (separately) rare earths — requiring licences from 8 November. For Indian cell-makers dependent on Chinese equipment, vendor ecosystems, and licensed IP, this is a live supply-chain and technology-transfer risk (already manifest in PLI delays from visa hold-ups for Chinese engineers).

6. Regional policy — the protection question

Table 5 — Regional policy and competitiveness

Region	Key policy	Direction 2025–26	Implication
China	Industrial policy, "anti-involution," export controls	Consolidating; forcing utilisation >60%	Retains cost + supply-chain dominance
Europe	EU Battery Regulation, CBAM, local content	Support skewed; Northvolt failed	Import dependence on Asia rising
US	IRA 45X/30D → OBBA (Jul 4 2025)	30D EV credit killed after 30 Sep 2025; 45X phased, FEOC restrictions	Cell manufacturing credit survives but consumer pull cut; FEOC blocks Chinese-linked projects
India	ACC PLI (Rs18,100 cr), FAME/PM E-	PLI faltering (2.8%);	Demand support real; supply

	DRIVE, BESS VGF, ALMM	BESS VGF strong	localisation stalled
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The US **One Big Beautiful Bill Act** terminated the 30D/45W clean-vehicle credits after 30 September 2025 and imposed FEOC restrictions on six credits (45X, 45Y, 48E, 45Q, 45Z, 45U) — a material negative for US EV demand and for any project with Chinese content. India, by contrast, is layering demand support (GST on EVs cut to 5%; BESS VGF approved for 43.8 GWh) but has failed to localise cells.

7. India deep dive

Demand. EV penetration reached c.7.3% of vehicles in Q3 FY26, with 2W/3W c.87% of EV volumes; electric-car penetration is still low single digits. NITI Aayog targets 30% EV sales by 2030 (80% 2W/3W, 30% private cars, 70% commercial cars, 40% buses), but its own September 2025 report concedes India must lift the EV share by 22 points in five years to hit target — a stretch. Lead-acid, meanwhile, keeps growing: CRISIL sees the lead-acid industry up 10–11% in FY25, driven by autos, 5G/telecom, data centres and railways.

Cell localisation — announced vs real.

Table 6 — India cell capacity tracker (announced vs commissioned)

Player	Announced GWh	Partner/chem	Status	A/E
Ola Electric	5 (of 20 by FY29)	In-house 4680 NMC	1.4 GWh commissioned — only PLI delivery	A
Reliance New Energy	30–120	Faradion (Na-ion) IP	2 GWh Na-ion Phase-1 (Sep 2025)	A/E
Tata Agratas	20 (Sanand)	Envision AESC	Under construction, ops c.2028	E
JSW	10→50	LG Energy Solution	MoU/planning	E
Exide (EESL)	12 (2 phases)	SVOLT	Phase-1 6 GWh, samples Q1FY27	A/E
Amara Raja (ARACT)	16	Gotion/GIB (LFP), Chinese NMC	CQP live; 2 GWh cells by Jun-2027	A/E

Others (Godi, HBL, Lucas TVS, Livguard, Adani)	60+	Various	Mostly announced	E
India total intent by 2030	~290 GWh	—	vs 1.4 GWh live today	E

Judgement: the gap between c.290 GWh of intent and 1.4 GWh delivered is the entire story. Near term (to c.2027) India stays import-dependent — Nuvama flags a c.210 GWh shortfall vs the 2032 BESS requirement against a c.68 GWh pipeline. Medium term (2028–30), if Reliance/Tata/JSW tonnage lands alongside Chinese import competition at \$50–70/kWh packs, India faces the same *oversupply* fate as China and Europe unless import duties protect domestic cells. There is no scenario in which all 290 GWh runs profitably at scale.

BESS — the credible pull. 2025 saw 130+ GWh tendered; 10.4 GWh of standalone BESS allocated. Tariffs collapsed: standalone 2-hour bids reached **Rs1.48 lakh/MW/month** (APTRANSCO, Dec 2025), with 4-hour standalone clearing Rs6.45–6.46/kWh (UPPCL, May 2026). VGF moved from Rs27 lakh/MWh (Tranche I) to Rs18 lakh/MWh (Tranche II, Jul 2025). CEA/NEP require **8.68 GW/34.7 GWh BESS by FY27 and 47.24 GW/236 GWh by FY32** (411 GWh total storage incl. PSP), rising to 1,840 GWh BESS by 2047. This is the demand pool most aligned with Exide/Amara Raja's LFP ambitions — but IEEFA warns c.75% of allocated 2-hour capacity was bid below viable tariff benchmarks, so developer economics (and hence cell pricing power) are thin.

Recycling. Battery Waste Management Rules 2022 (2025 amendments) mandate rising recovery (90% by FY27), 70% EV-battery collection by FY28, and domestic recycled content (5% from FY28 → 20% by FY31). LOHUM (c.70% share), Attero, Recyclekaro and LICO are the registered leaders. NITI Aayog estimates recovered materials could meet 35% of domestic lithium/nickel demand by 2030 — but only c.1% of batteries are formally recycled today. A structural opportunity, but not yet material to Exide/Amara Raja earnings.

PLI reality. Rs18,100 cr outlay; 50 GWh target by 2025; **1.4 GWh delivered (2.8%)** by October 2025, all Ola, with zero incentives disbursed against a targeted Rs29 bn and just 1,118 jobs created vs a 1.03 mn target (IEEFA/JMK, Jan 2026). Hyundai Global withdrew its 20 GWh; Reliance, Ola and Rajesh Exports sought one-year extensions and face penalties (0.1%/day of performance security — Rs12.5 lakh/day for Ola, Rs5 lakh/day each for Reliance and Rajesh). Neither Exide nor Amara Raja is a core PLI beneficiary — they are self-funding, which insulates them from PLI penalties but denies them the subsidy cushion.

8. The two stocks — financials and valuation

Table 7 — Exide vs Amara Raja (consolidated, FY26 unless noted)

Metric	Exide (EXID IN)	Amara Raja (ARENMOB IN)
FY26 revenue	c.Rs17,269 cr	Rs13,814 cr
FY26 EBITDA	c.Rs1,943 cr	c.Rs1,544 cr
Core EBITDA margin	c.11.2%	c.10.8% (LAB EBIT 13.2% Q4)
FY26 PAT	c.Rs1,111 cr (std) / Rs795 cr FY25 cons	Rs895.8 cr
Lithium subsidiary	EESL: FY26 loss Rs248 cr; Rs4,902 cr infused	ARACT: c.Rs1,500 cr infused
Lithium plan	12 GWh, ~Rs6,000 cr, SVOLT	16 GWh, Rs9,500 cr, Gotion
First cell revenue	Q3 FY27 (guidance)	H2 FY27 (2 GWh Jun-2027) Sahi
Net debt	Zero (net cash)	Net-cash; CRISIL AA+
Dividend FY26	Rs2.0/sh	Rs10.60/sh
Market cap	c.Rs33,000–37,700 cr	c.Rs16,000–18,200 cr
P/E	c.28x (39.9x cons, EESL drag)	c.21x
Consolidated ROE	c.6%	c.11%
Street rating	Hold/Neutral (Motilal 327, ICICI 480, Nomura 404)	Hold/Buy (ICICI 960 HOLD, Nuvama 1,120 Buy)

How the street values the lithium optionality. This is the crux of both stocks. Analysts value the **core lead-acid at c.15–18x forward EPS** and add the **lithium business at roughly invested capital / book — not a growth premium:**

- **Exide:** Motilal Oswal ascribes Rs59/share to lithium (book); ICICI Securities Rs60/share (1x invested capital); both value core at 15–18x FY28E. [Investment Guru](#) Nomura (Neutral, Rs404) warns “unless import duties imposed, margins/ROCEs likely subdued.” [business-standard](#)
- **Amara Raja:** Nuvama (Buy, Rs1,120) ascribes Rs169/share to lithium (1x P/B) plus Rs42/share investments, core at 15x; [business-standard](#) ICICI Securities (HOLD, Rs960)

uses 0.8x CWIP on Li-ion capex. ICICI Direct

The debate in one line: the market is telling you the cell business is worth what has been spent on it and no more. That is the correct default until utilisation, offtake and returns are proven.

Investment framework — what has to be true for the cell bet to create value

Distinguish structural from cyclical. Own these names for the *structural* lead-acid replacement franchise (autos, UPS/inverter, telecom, data-centre standby) plus a *free-ish option* on lithium — not for a lithium growth story you pay up for. The lead-acid cash cow is the thesis; lithium is the call option.

The consensus vs our view. *Consensus:* both are “legacy-to-EV transformation” stories where in-house cells re-rate the multiple. *Our differentiated view:* absent protective import duties and locked offtake, Indian cells will be sub-scale, technology-dependent, and squeezed between Chinese imports at \$50–70/kWh and domestic giants (Reliance, Tata) with deeper pockets — the base case is value-neutral to value-destructive on the cell capex, exactly as the street’s at-cost valuation implies. The re-rate is a bull case, not a base case.

Five conditions for the cell business to create rather than destroy value:

1. **Protection:** meaningful customs duty / non-tariff barrier on imported cells (today India is c.100% import-dependent; no such duty yet).
2. **Offtake:** binding, priced contracts (Amara Raja has Ather/Piaggio MoUs; Exide has a Hyundai/Kia LFP MoU — all non-binding so far).
3. **Utilisation:** cross breakeven (Amara Raja: 2 GWh breakeven, 4 GWh for profit) business-standard — watch the ramp from FY27.
4. **Cost:** land within c.20–30% of Chinese cell cost via SVOLT/Gotion process transfer, despite lacking local CAM/AAM.
5. **Returns:** achieve asset turns $\geq 1x$ and ROCE above the c.11–14% the core lead-acid already earns — otherwise the capex dilutes group returns (already visible: consolidated ROE dragged to c.6% at Exide).

Bull / Base / Bear scenarios for the lithium leg

Scenario	Assumptions	Implication

Bull	Import duty on cells; binding OEM/BESS offtake; utilisation >60%; positive cell gross margin by FY28	Re-rate: lithium worth >1x invested capital, SOTP uplift Rs100–200/sh
Base	Slow ramp, thin margins, no duty, cells at import parity	Lithium $\approx$ invested capital (street's current stance); no re-rate
Bear	Sodium-ion/Chinese imports undercut Indian LFP; utilisation <40%; capex overruns	Write-down risk; lithium a drag on group ROCE and cash

Leading indicators to monitor (trackable series):

- **Vahan** monthly EV registrations by segment (2W/3W the leading edge).
- **BNEF pack/cell price index** and **China LFP cell \$/kWh** (the import-parity threat).
- **Lithium carbonate spot (GFEX)** and **sodium-ion cost curve** (chemistry-substitution risk).
- **BESS tender tariffs** (Rs/MW/month, Rs/kWh) and **VGF tranches** (demand-pool economics).
- **PLI milestone disclosures** and **penalty notices** (ecosystem health).
- **EESL / ARACT quarterly losses, capex infusions, and first-revenue disclosures** (the ramp).
- **India cell import data** (customs) and any **import-duty / ALMM-type notification** (the single biggest swing factor).
- **Antimony and lead prices** (near-term lead-acid margin driver — spiked in FY25).

Recommendations

Stage 1 — Now (own the cash cow, not the option). Prefer the name where the *core* is cheaper and returns higher relative to the lithium drag. Amara Raja screens cheaper (c.21x, ROE c.11%, LAB EBIT 13.2%, Rs10.6 dividend) with a lighter relative lithium cash-burn; Exide's consolidated optics are worse (ROE c.6%, EESL Rs248 cr loss) but its 12 GWh is further along (samples Q1FY27). Treat both as lead-acid compounders with a lithium option priced at cost — a reasonable risk/reward, not a re-rating trade.

Stage 2 — Re-rate trigger (upgrade the option). Turn constructive on the lithium leg only when you see: (a) an import-duty/ALMM notification on cells, OR (b) a binding, priced multi-

GWh offtake with a named OEM/BESS developer, OR (c) utilisation crossing breakeven with disclosed positive cell gross margin. Any one materially de-risks the SOTP's at-cost lithium stub.

Stage 3 — De-rate/avoid triggers. Cut the lithium optionality to zero (or negative) if: sodium-ion undercuts Indian LFP economics in 2W/3W/BESS before FY28; Chinese import packs sustain <\$60/kWh into India with no duty; or EESL/ARACT capex escalates beyond plan with slipping first-revenue dates. Also watch lead/antimony spikes compressing the core margin that funds everything.

Benchmarks that change the call: import duty on cells (binary positive); BESS tariffs stabilising above IEEFA viability benchmarks (positive for LFP pricing power); cell utilisation >60% at either plant (positive); Chinese pack imports <\$60/kWh sustained with no duty (negative).

Caveats

- **Actuals vs estimates:** 2024–25 prices, market shares and BESS tariffs are actuals; all 2030+ demand/capacity figures and company first-revenue dates are estimates or company guidance and should be treated as such.
- **Company financials carry reporting ambiguity:** Exide's FY26 headline revenue/EBITDA/PAT are variously described as standalone vs consolidated across sources; EESL losses consolidate separately. Market-cap and P/E figures differ by source and date (Exide c.Rs33,000–37,700 cr; P/E 28x vs 39.9x depending on EESL drag). Verify against company results PDFs before modelling.
- **Street SOTP figures** (Exide lithium Rs59–60/share; Amara Raja Rs169/share) are quoted from broker notes of differing vintages (Motilal, ICICI Securities, Nuvama, Nomura); pull the primary PDFs for verbatim tables.
- **Forward-looking claims flagged:** CATL sodium timelines, Toyota solid-state dates, India 290 GWh and NITI 30% targets, and FY28 revenue aspirations (Exide's Rs20,000 cr) are announcements/projections, not banked outcomes.
- **Contested data:** India's "credible" cell demand vs 290 GWh intent is our judgement, not a consensus number; BESS demand ranges widely across CEA scenarios (193–335 GWh by FY32).